Cluster Analysis in Unsupervised Learning

1. Introduction to Cluster Analysis

Cluster analysis is a powerful technique in **unsupervised learning**, where the goal is to identify patterns or structures in data without any predefined labels. Unlike supervised learning—where the model learns from labeled examples—unsupervised learning aims to discover hidden relationships within the dataset itself.

Clustering divides a dataset into groups, called **clusters**, in which data points within a cluster are more **similar to each other** than to those in other clusters. Each cluster represents a collection of data items that share common properties or features.

For example, in a retail dataset, clustering can identify customer groups based on purchasing behavior—such as budget shoppers, luxury buyers, or occasional customers. Similarly, in biology, clustering can group genes with similar expression patterns.

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2. Clustering in Unsupervised Learning

In **unsupervised learning**, we do not provide the algorithm with predefined labels or outcomes. The system observes data and finds patterns purely based on **similarity** and **structure**.

Key Objectives of Clustering:

- To identify **natural groupings** in data.
- To summarize data for easier understanding.
- To find **anomalies or outliers** that do not fit any group.
- To provide preprocessing for other algorithms like classification.

Applications of Clustering:

- Market segmentation.
- Image compression.
- Document or text grouping.
- Customer behavior analysis.

3. Partition Criteria: Dividing Data Points

Clustering algorithms partition a dataset into groups based on specific **criteria**. The **partition criterion** defines how clusters are formed and what qualifies data points to belong to the same or different clusters.

Types of Partitioning Criteria:

1. **Compactness Criterion:**
   Clusters should be compact—data points within a cluster should be close to the cluster's centroid.
   Example: K-Means uses the **sum of squared distances** between points and their centroids as the objective function.

2. **Separation Criterion:**
   Clusters should be well-separated—points in different clusters should be far apart.
   Example: Algorithms may try to maximize the **distance between cluster centroids**.

3. **Connectivity Criterion:**
   Points that are close in the input space should belong to the same cluster.
   Example: Hierarchical and density-based clustering methods often use this idea.

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The partitioning process is iterative, where data points are reassigned to clusters until an optimal configuration is achieved based on the chosen criterion.

4. Exclusive or Non-Exclusive Data

When forming clusters, the nature of data assignment to clusters can be **exclusive** or **non-exclusive**.

a. Exclusive (Hard) Clustering:

Each data point belongs to **exactly one cluster**.

- Example: K-Means assigns each point to the nearest centroid.
- Pros: Simple and easy to interpret.
- Cons: Fails when data points lie between clusters or overlap.

b. Non-Exclusive (Soft/Fuzzy) Clustering:

A data point can belong to **multiple clusters** with varying degrees of membership.

- Example: Fuzzy C-Means algorithm.
- Each data point has a **membership value (0 to 1)** representing its association with each cluster.
- Pros: Handles overlapping clusters better.
- Cons: Computationally more complex.

c. Overlapping vs Non-Overlapping:

- **Overlapping:** Data points can exist in multiple clusters. (e.g., a movie can belong to both "comedy" and "romantic" clusters)
- **Non-overlapping:** Each point strictly belongs to one cluster.

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5. Similarity Measure

Similarity measures are crucial in clustering because they define how alike two data points are. The **more similar** two objects are, the **closer** they should be in the cluster space.

Similarity can be calculated in several ways depending on the type of data (numerical, categorical, binary, or mixed).

a. For Numerical Data:

Euclidean Distance:

$$d(x,y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$$

Most commonly used;measures straight-line distance in n-dimensional space.

Manhattan Distance (L1 norm):

$$d(x, y) = \sum_{i=1}^{n} |x_i - y_i|$$

Measures distance along grid-like paths (useful for sparse data)

Minkowski Distance:

A generalization of both Euclidean and Manhattan distances:

$$d(x, y) = (\sum |x_i - y_i|^p)^{1/p}$$

b. For Categorical Data:

Simple Matching Coefficient (SMC):

$$SMC = \frac{\text{Number of matches}}{\text{Total attributes}}$$

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Jaccard Coefficient:

Used when only positive matches matter (e.g., presence of features):

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

c. For Mixed Data:

Gower's Coefficient is often used as it handles numerical, categorical, and binary attributes simultaneously.

6. Distance-Based Similarity Measure

In clustering, the **distance-based similarity measure** quantifies how far apart two data points are in feature space.
Smaller distances mean higher similarity.

Common Distance Measures:

1.Euclidean Distance – Suitable for continuous, normalized data.

2.Cosine Similarity – Often used in text mining:

$$\text{Cosine Similarity} = \frac{x \cdot y}{||x||\,||y||}$$

Measures the angle between two vectors rather than the distance.

3.Hamming Distance – Counts differences between binary strings.

4.Mahalanobis Distance – Accounts for correlation between variables and the shape of data distribution.

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Normalization:

Before applying distance-based measures, data should be **standardized** to avoid dominance of variables with large scales.

7. Clustering Space

The **clustering space** refers to the multidimensional space where each data point is represented as a vector of features. Clusters are formed as regions of high data density separated by regions of low density.

Concept:

- Each dimension corresponds to one feature.
- Data points are plotted based on their feature values.
- The clustering algorithm identifies regions with similar patterns.

Example:

In a 3D clustering space representing students:

- X-axis: Study hours
- Y-axis: Grades
- Z-axis: Attendance
  Clusters may reveal “high achievers,” “average performers,” and “low-attendance students.”

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Visualization tools like PCA (Principal Component Analysis) can help reduce dimensions to visualize clusters effectively.

8. Hierarchical Clustering

Hierarchical clustering builds a hierarchy of clusters either **from bottom-up (agglomerative)** or **top-down (divisive)**.

a. Agglomerative Hierarchical Clustering (AHC):

- Starts with each data point as its own cluster.
- Iteratively merges the two most similar clusters.
- Continues until all data points are in a single cluster.

b. Divisive Hierarchical Clustering:

- Starts with one large cluster containing all data points.
- Iteratively splits clusters into smaller ones.

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Linkage Methods:

These define how distances between clusters are computed:

1. **Single Linkage:** Minimum distance between cluster points.
2. **Complete Linkage:** Maximum distance between cluster points.
3. **Average Linkage:** Average distance between all pairs of points.
4. **Ward's Method:** Minimizes variance within clusters.

Advantages:

- No need to predefine the number of clusters.
- Produces a visual structure (Dendrogram).

Disadvantages:

- Computationally expensive ($O(n^3)$).
- Sensitive to noise and outliers.

9. Dendrogram

A **dendrogram** is a tree-like diagram that illustrates the arrangement of clusters produced by hierarchical clustering.

Features:

- Each **leaf node** represents an individual data point.
- Each **branch** represents the merging of clusters.
- The **height** of the branch indicates the distance (dissimilarity) at which clusters were merged.

Interpreting Dendrograms:

- The vertical axis shows dissimilarity (distance).
- Cutting the dendrogram at a chosen height yields the desired number of clusters.
- Lower merge heights indicate more similar clusters.

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Example:

In customer segmentation, a dendrogram can show how groups of similar customers merge as similarity thresholds decrease.

10. K-Means Algorithm

The **K-Means algorithm** is one of the most popular partition-based clustering methods. It divides data into **K non-overlapping clusters** based on minimizing the **within-cluster variance**.

Steps:

1. Choose the number of clusters K.

2.Randomly initialize K centroids.

3.Assign each data point to the nearest centroid.

4.Recalculate centroids as the mean of points in each cluster.

5.Repeat steps 3–4 until centroids stabilize or convergence is achieved.

Objective Function:

$$J = \sum_{i=1}^{K} \sum_{x_j \in C_i} ||x_j - \mu_i||^2$$

Where μi is the centroid of cluster Ci.

Advantages:

- Simple and efficient for large datasets.
- Works well when clusters are spherical and balanced.

Disadvantages:

- Must predefine K.
- Sensitive to initialization and outliers.
- Assumes clusters of similar size and density.

Example:

Grouping students based on marks and attendance into “high,” “medium,” and “low” performers.

11. K-Medoids Algorithm

The **K-Medoids algorithm** (or **PAM – Partitioning Around Medoids**) is similar to K-Means but uses **medoids** (actual data points) instead of centroids.

Steps:

1. Choose K random medoids.
2. Assign each data point to the nearest medoid.
3. Compute the total cost (sum of dissimilarities).
4. Replace medoids with non-medoid points if it reduces the total cost.
5. Repeat until no change occurs.

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Advantages:

- More robust to outliers because medoids are real data points.
- Works with any distance metric.

Disadvantages:

- Slower for large datasets compared to K-Means.

Example:

In customer data, a medoid represents an actual customer whose behavior is most typical of that cluster.

12. CLARA (Clustering Large Applications)

CLARA is an extension of K-Medoids designed to handle **large datasets** efficiently.

How it Works:

1. Draw multiple random samples of the dataset.
2. Apply the **PAM (K-Medoids)** algorithm on each sample.
3. Evaluate the clustering quality on the entire dataset.
4. Choose the clustering with the best quality (lowest cost).

Advantages:

- Scalable to large datasets.
- Maintains robustness of K-Medoids.

Disadvantages:

- Sampling may miss some patterns in data.
- Performance depends on sample size.

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Example:

Used for clustering millions of transactions to identify customer spending segments efficiently.

13. Grid-Based Approach

Grid-based clustering divides the data space into a finite number of cells that form a **grid structure**. Clusters are formed based on the **density of points** within these cells.

Steps:

1. Quantize the data space into a grid of cells.
2. Count the number of points in each cell.
3. Identify dense cells that exceed a certain threshold.
4. Merge adjacent dense cells to form clusters.

Common Grid-Based Algorithms:

- **STING (Statistical Information Grid):** Uses statistical info stored in each grid cell.
- **WaveCluster:** Uses wavelet transforms to detect dense regions.
- **CLIQUE:** Handles high-dimensional data using subspace clustering.

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Advantages:

- Fast processing time (depends on number of cells, not data points).
- Suitable for high-dimensional data.
- Easy to update clusters dynamically.

Disadvantages:

- Requires choosing grid size carefully.
- May not perform well if clusters are not aligned with grid boundaries.

Example:

Used in spatial data mining (e.g., grouping geographic locations based on population density).

14.Conclusion

Cluster analysis is a cornerstone of unsupervised learning. It helps uncover hidden patterns, relationships, and structures in data that would otherwise remain unseen. From simple algorithms like K-Means to advanced techniques like CLARA and Grid-based clustering, each approach offers unique strengths depending on the dataset and application.

Key insights:

- Similarity and distance measures define how clusters are formed.
- Partition and hierarchical methods serve different purposes—partitioning for efficiency and hierarchy for interpretability.
- Modern clustering techniques increasingly combine statistical, density-based, and spatial insights to handle large-scale, real-world data.

In the world of AI and data science, clustering remains an essential step in data exploration, feature extraction, and knowledge discovery.

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